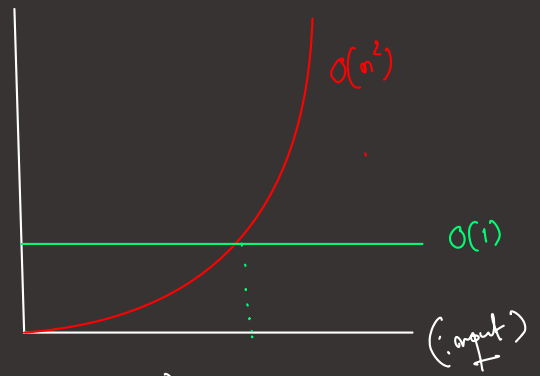


$$f(n) = an^2 + n \log n + c \rightarrow O(n^2)$$

Ω
Lower bound

O
Upper bound



$$\underbrace{\log n < 4n + 10}_{\text{lower bound}} < f(n) = 5n + 10 < \underbrace{6n + 10 < n \log n < n^2}_{\text{upper bound}}$$

$\downarrow$
 $\Theta(n)$

Annotations: $\Omega(n)$ and $O(n)$ are written above the lower bound, and $\Theta(n)$ and $O(n)$ are written above the upper bound.

$$\Omega \leq \Theta \leq O$$


```

for (i=0; i < n; i++) {
    for (j=i+1; j < n; j++) {
        start;
    }
}

```

$i=0$, $j=1$ to $(n-1)$, start $\rightarrow (n-1)$
 $i=1$, $j=2$ to $(n-1)$, start $\rightarrow (n-2)$
 $i=2$, $j=3$ to $(n-1)$, start $\rightarrow (n-3)$
 $\vdots$
 $i=(n-2)$, $j=n-1$ to $n-1$, start $\rightarrow 1$
 $i=(n-1)$, $j=n$ $\times$ start $\rightarrow 0$
 $i=n$ $\times$

$$\therefore \text{time } f_n^n: 1 + 2 + \dots + (n-1)$$

$$= \frac{n(n-1)}{2}$$

$$\therefore \text{C.C.} = O(n^2)$$

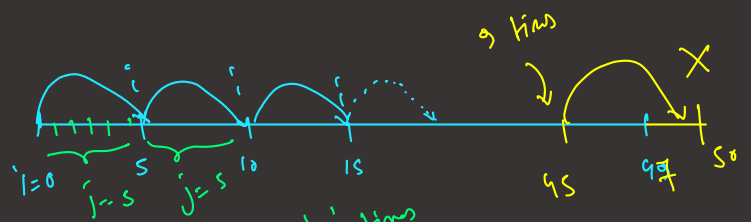
assume $k < n$

```

for (i=0; i < n; i=i+k) {
    for (j=1; j <= k; j++) {
        start
    }
}

```

$n=48$
 $k=5$



for each sub-loop inner loop runs 'k' times

$\therefore$ time f_n^n :

$$\frac{n}{k} \cdot k = n$$

$$\therefore O(n)$$

How many times this outer loop is running?

$$\left\lfloor \frac{n}{k} \right\rfloor \text{ times?}$$